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1. Ders

Nişantepe Mah.
Orman Sok. No:34-36
34794 Alemdağ
Çekmeköy İstanbul
T: 0216 564 9000
F: 0216 564 9999
info@ozyegin.edu.tr
www.ozyegin.edu.tr

Invariance of Scalar Product Under Rotational Transformation:

By definition, scalar product proportional to $|\vec{A}|, |\vec{B}|, \cos \theta_{AB}$. None of these changes by the coordinate system, thus scalar product, which is a scalar quantity is expected to be the same.

$\vec{A} \cdot \vec{B} = \vec{B} \cdot \vec{A}$ where $\vec{A}$ representation of the vector in the rotated coord sys

$$\vec{A}' \cdot \vec{B}' = \sum_k A'_k \cdot B'_k$$

$$= \underbrace{\sum_i a_{xi} A_i}_{A'_x} \underbrace{\sum_j a_{xj} B_j}_{B'_x} + \underbrace{\sum_i a_{yi} A_i}_{A'_y} \underbrace{\sum_j a_{yj} B_j}_{B'_y} + \underbrace{\sum_i a_{zi} A_i}_{A'_z} \underbrace{\sum_j a_{zj} B_j}_{B'_z}$$

$$= \sum_l \left(\sum_i a_{li} A_i \cdot \sum_j a_{lj} B_j \right) = \sum_l \sum_i \sum_j a_{li} a_{lj} A_i \cdot B_j$$

$$= \sum_i \sum_j \underbrace{\left(\sum_l a_{li} a_{lj} \right)}_{\delta_{ij}} A_i B_j$$

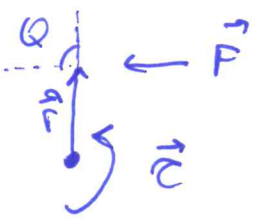
only the terms when $i=j$ survives.

$j \rightarrow i$

$$\sum_i A_i B_i = A' \cdot B'$$

$$A' \cdot B' = \sum_i \sum_j \delta_{ij} A_i B_j$$

Vector Product (Cross Product).



$\vec{C}$ is proportional to $|\vec{F}|$, $|\vec{r}|$ and $\sin Q$

$\vec{C}$ is vector, and direction is both $\perp \vec{F}$ and $\vec{r}$, and obeys R.H.R.

For notation

$$\vec{C} = \vec{A} \times \vec{B}, \quad |\vec{C}| = |\vec{A}| \cdot |\vec{B}| \cdot \sin \theta_{A \rightarrow B}$$

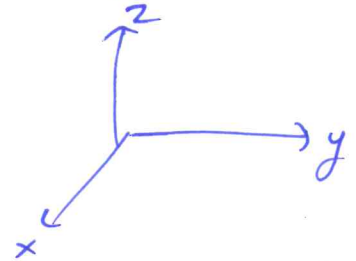
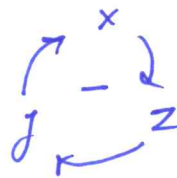
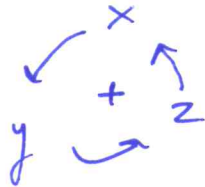
To satisfy RHR $\vec{A} \times \vec{B} = -\vec{B} \times \vec{A}$ ← anti commuting.

$$\hat{x} \times \hat{x} = \hat{y} \times \hat{y} = \hat{z} \times \hat{z} = 0$$

$$\hat{x} \times \hat{y} = \hat{z}$$

$$\hat{y} \times \hat{z} = \hat{x}$$

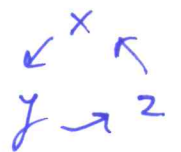
$$\hat{z} \times \hat{x} = \hat{y}$$



$$\vec{C} = \vec{A} \times \vec{B} = \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ A_x & A_y & A_z \\ B_x & B_y & B_z \end{vmatrix} = \hat{x} \begin{vmatrix} A_y & A_z \\ B_y & B_z \end{vmatrix} - \hat{y} \begin{vmatrix} A_x & A_z \\ B_x & B_z \end{vmatrix} + \hat{z} \begin{vmatrix} A_x & A_y \\ B_x & B_y \end{vmatrix}$$

Due to symmetry between x, y, z we may use Levi-Civita symbol

$$\epsilon_{ijk} = \begin{cases} +1 & \text{if } i, j, k \text{ cyclic } \curvearrowright \text{ direction} \\ 0 & \text{if any two or three indices repeat} \\ -1 & \text{if } i, j, k \text{ cyclic } \curvearrowleft \text{ direction} \end{cases}$$



$$\epsilon_{123} = \epsilon_{231} = \epsilon_{312} = 1$$

$$\epsilon_{321} = \epsilon_{213} = \epsilon_{132} = -1$$

$$\epsilon_{112} = \epsilon_{111} = \dots = 0.$$

$$\vec{C} = \vec{A} \times \vec{B} \Rightarrow C_i = \sum_j \sum_k \epsilon_{ijk} A_j B_k = \epsilon_{ijk} A_j B_k.$$

Ex 1 Show that $\vec{D} = (\vec{A} \times \vec{B}) \cdot \vec{C} \Rightarrow \vec{D} = \epsilon_{ijk} A_j B_k C_i$

$\wedge$
 implied $\sum_i \sum_j \sum_k$

Properties of Levi-Civita (3D)

$$1) \sum_i \epsilon_{ijk} \epsilon_{imn} = \delta_{jm} \delta_{kn} - \delta_{jn} \delta_{km}$$

$$2) \sum_i \sum_j \epsilon_{ijk} \epsilon_{ijn} = 2 \delta_{kn}$$

$$3) \text{ Note that } \epsilon_{ijk} = \epsilon_{jki} \text{ and } \epsilon_{ijk} = -\epsilon_{jik}$$

Ex 2 (HW)

Show that $[\vec{A} \times (\vec{B} \times \vec{C})]_i = [\vec{B}(\vec{A} \cdot \vec{C}) - \vec{C}(\vec{A} \cdot \vec{B})]_i$

Q, $\vec{A} \cdot (\vec{A} \times \vec{B}) = ?$ show that with indicies.

Q.

Q, ~~$\vec{A} \cdot (\vec{B} \times \vec{C})$~~ $D = \vec{A} \cdot (\vec{B} \times \vec{C}) = \begin{vmatrix} x & y & z \\ B_x & B_y & B_z \\ C_x & C_y & C_z \end{vmatrix}$ $\xrightarrow{\text{show with indicies}}$